

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

- a) Write a C code to create a process using fork() system call.
- b) Read the user input as shell command.
- c) Use a system call exec() to execute the shell command.
- e) Display the PIDs of parent and child processes.

```
root@Pranay:~# nano Task1.c
root@Pranay:~#
```

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

#define MAX_INPUT 100
#define MAX_ARGS 20

int main() {
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    printf("Enter a Linux command: ");
    fgets(input, sizeof(input), stdin);

    // Remove newline character
    input[strcspn(input, "\n")] = '\0';

    // Split the command into arguments
    int i = 0;
    args[i] = strtok(input, " ");

    while (args[i] != NULL && i < MAX_ARGS - 1) {
        i++;
        args[i] = strtok(NULL, " ");
    }

    pid_t pid = fork();

    if (pid < 0) {
        // Fork failed
        printf("Fork failed!\n");
    }
}
```

```

        return 1;
    }
    else if (pid == 0) {
        // Child process
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        // Execute command
        execvp(args[0], args);

        // Executes only if exec fails
        perror("Command execution failed");
        exit(1);
    }
    else {
        // Parent process
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID : %d\n", pid);
    }
}

```

1)Then press ctrl+o enter ctrl+x2)then gcc filename.c -o filename

```

root@Pranay:~# nano Task1.c
root@Pranay:~# gcc Task1.c -o Task1
root@Pranay:~#

```

3)then ./filename

```

root@Pranay:~# nano Task1.c
root@Pranay:~# gcc Task1.c -o Task1
root@Pranay:~# ./Task1
Enter a Linux command:

```

4)enter the linux command

```

root@Pranay:~# ./Task1
Enter a Linux command: ls

Parent Process

Child Process

```

```

Parent PID: 474
Child PID : 477
Child PID : 477
Parent PID: 474
Dashboard    LICENSE    OSSP    Task1    Task1.c.save    include    shopping.c    skill
src
Dashboard.c  Makefile    README.md Task1.c filename.c    shopping    shopping.c.save skill.c

Child process completed.
root@Pranay:~#

```

Child process completed.

Task 2

Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

An operating system is system software that controls all the hardware and software resources of a computer. Users do not communicate directly with the hardware. Instead, the operating system acts as a bridge between the user, applications, and the hardware

Linux Commands Used

1. uname

The uname command is used to display information about the Linux operating system. It provides details such as the operating system name, kernel version, machine architecture, and hostname. This command helps us identify the system on which Linux is running.

```

root@Pranay:~# uname
Linux
root@Pranay:~#

```

2. lscpu

The lscpu command displays detailed information about the processor. It shows the CPU model, number of cores, number of threads, CPU architecture, and processor speed. This information helps us understand the hardware capabilities of the computer.

```

root@Pranay:~# lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12

```

```

On-line CPU(s) list:   0-11
Vendor ID:             GenuineIntel
Model name:            12th Gen Intel(R) Core(TM) i5-1235U
CPU family:            6
Model:                 154
Thread(s) per core:    2
Core(s) per socket:    6
Socket(s):              1
Stepping:               4
BogoMIPS:              4992.00
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat
pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good
nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx sse3 fma cx16
pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand
hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid
ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb
sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnmi umip waitpkg gfni vaes vpclmulqdq rdpid
movdiri movdir64b fsrm md_clear serialize ibt flush_l1d arch_capabilities
Virtualization features:
Virtualization:        VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                    288 KiB (6 instances)

```

3. ps

The `ps` command lists the processes that are currently running in the system. It displays the Process ID (PID), Parent Process ID (PPID), user name, and the command associated with each process. This command is useful for monitoring and managing processes.

```

root@Pranay:~# ps
  PID TTY          TIME CMD
  435 pts/0    00:00:00 bash
  503 pts/0    00:00:00 ps
root@Pranay:~#

```

4. top

The `top` command provides a real-time view of the system's performance. It continuously displays CPU usage, memory usage, running processes, and system load. It is commonly used to identify programs that consume a large amount of system resources.

```

top - 05:44:46 up 11 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks: 22 total,   1 running, 21 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st

```

MiB Mem : 7804.6 total, 7276.2 free, 522.0 used, 156.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7282.7 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21532	13044	9724	S	0.0	0.2	0:00.53	systemd
2	root	20	0	3180	2200	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	50448	16000	14840	S	0.0	0.2	0:00.29	systemd-journal
97	root	20	0	25380	6788	5168	S	0.0	0.1	0:00.26	systemd-udev
165	systemd+	20	0	21344	13232	11108	S	0.0	0.2	0:00.10	systemd-resolve
166	systemd+	20	0	91036	7988	7020	S	0.0	0.1	0:00.08	systemd-timesyn
172	root	20	0	4244	2680	2428	S	0.0	0.0	0:00.00	cron
173	message+	20	0	9528	5308	4612	S	0.0	0.1	0:00.06	dbus-daemon
180	root	20	0	17980	8856	7812	S	0.0	0.1	0:00.05	systemd-logind
186	root	20	0	1756104	13112	10888	S	0.0	0.2	0:00.10	wsl-pro-service
193	syslog	20	0	222516	5988	4624	S	0.0	0.1	0:00.07	rsyslogd
201	root	20	0	3124	1976	1836	S	0.0	0.0	0:00.00	agetty
213	root	20	0	107032	23288	13836	S	0.0	0.3	0:00.10	unattended-upgr
300	root	20	0	6672	4496	3728	S	0.0	0.1	0:00.01	login
343	root	20	0	20320	11580	9452	S	0.0	0.1	0:00.09	systemd
344	root	20	0	21168	3516	1804	S	0.0	0.0	0:00.00	(sd-pam)
368	root	20	0	6080	5276	3652	S	0.0	0.1	0:00.00	bash
430	root	20	0	3188	1112	980	S	0.0	0.0	0:00.00	SessionLeader
432	root	20	0	3204	1252	1108	S	0.0	0.0	0:00.07	Relay(435)
435	root	20	0	6080	5352	3672	S	0.0	0.1	0:00.03	bash
504	root	20	0	9280	5376	3224	R	0.0	0.1	0:00.00	top